Ops 5: Observability and Incident Response

Minecraft on Kubernetes (k3s) - CS 312 System Administration

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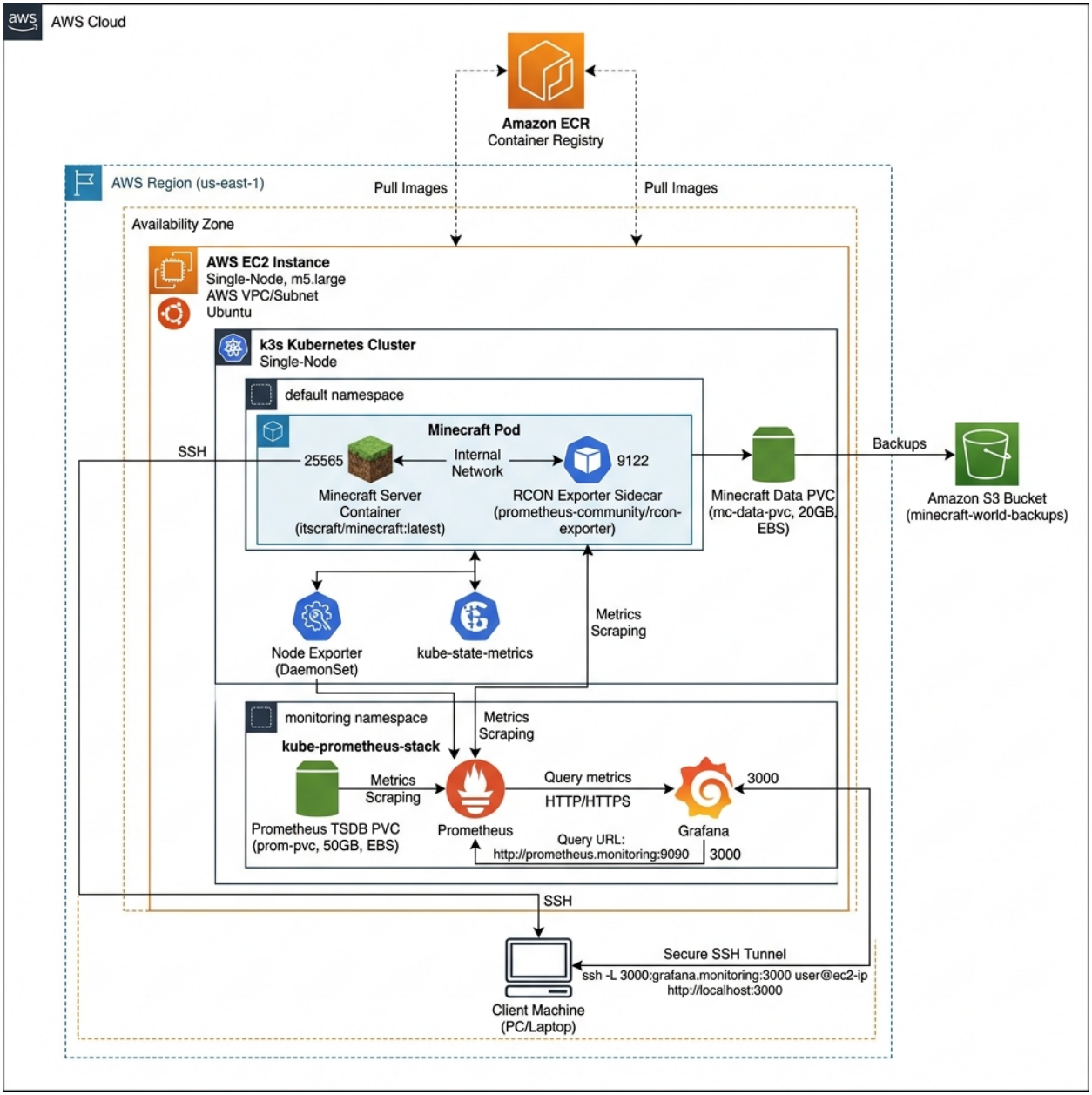
Repository	https://github.com/GenericProgram/cs312-minecraft-hws
Video	https://media.oregonstate.edu/media/t/1_g1abc3tt
Checkpoints	CP1: 00:03 | CP2: 00:38 | CP3: 01:10 | CP4: 02:07
AWS Region	us-east-1
Instance	t3.large, Ubuntu 22.04 LTS, 20 GB gp3
Kubernetes	k3s v1.35.5+k3s1 (single-node)
Image	998487032255.dkr.ecr.us-east-1.amazonaws.com/minecraft-server:v1.20.4

1. Architecture



The architecture builds upon the Ops 4 baseline, integrating the Minecraft server workload (with a `minecraft-exporter` sidecar) alongside the monitoring components in a dedicated `monitoring` namespace. The Minecraft server container image is pulled from a secure AWS ECR registry, and persistent world state backups are synchronized to an AWS S3 bucket. Prometheus scrapes host-level metrics via the Node Exporter daemonset, pod-level metrics via kube-state-metrics, and application-level metrics via the Minecraft exporter. Grafana provides visualization of these metrics, which is accessed securely via an SSH port-forward tunnel rather than exposing Grafana or Prometheus to the public web.

2. Repository File Map

All Ops 5 submission files are organized in the **ops3-4-5/** directory.

`helm/monitoring-values.yaml`	Declarative configurations, limits, and retention for kube-prometheus-stack.
`k8s/monitoring/namespace.yml`	Namespace isolation for the observability stack.
`k8s/monitoring/prometheus-rules.yml`	Declarative Prometheus alerts (CrashLoop, Mem/Disk pressure).
`ops-5-writeup/ops_5_writeup.pdf`	Self-contained PDF documentation including architecture, runbooks, and incident postmortem.
`playbook.yml`	Updated Ansible playbook triggering k3s, configuring ECR/S3, and applying manifests.
`k8s/deployment.yml`	Minecraft server deployment containing the primary server container, the RCON prometheus-exporter sidecar, and liveness/readiness/startup probes.
`k8s/configmap.yml`	Minecraft environment variables (pinned to version 1.20.4 to prevent Java 21 crash).
`k8s/service.yml`	Kubernetes Service exposing Minecraft TCP (25565) and RCON (25575) ports.
`k8s/pvc.yml`	Persistent Volume Claim mapping the Minecraft world-data directory to persistent host storage.
`k8s/rcon-secret.yml`	Kubernetes Secret storing the sensitive RCON password securely.
`k8s/backup-cronjob.yml`	Kubernetes CronJob for automated world-state synchronization to S3.
`terraform.tfvars`	Terraform variables reflecting the t3.large instance scaling.

3. Deployment Runbook

An operator can deploy the full monitoring stack onto the existing Ops 4 baseline by following these steps.

3.1 Prerequisites & Foundation Fixes

Ensure the EC2 instance is scaled to a `t3.large` via Terraform to accommodate the memory overhead of JVM, Prometheus, and Grafana simultaneously. The Minecraft ConfigMap must explicitly pin `VERSION: "1.20.4"` to prevent the container from fetching the latest version, which causes a Java 21 initialization crash.

3.2 Install Helm & Deploy Monitoring Stack

SSH into the EC2 instance and run the following to install Helm and the Prometheus stack declaratively:

```
curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | sudo bash

sudo KUBECONFIG=/etc/rancher/k3s/k3s.yaml helm repo add prometheus-community
https://prometheus-community.github.io/helm-charts
sudo KUBECONFIG=/etc/rancher/k3s/k3s.yaml helm repo update
```

```
sudo k3s kubectl create namespace monitoring

sudo KUBECONFIG=/etc/rancher/k3s/k3s.yaml helm install monitoring
prometheus-community/kube-prometheus-stack \
--namespace monitoring \
-f /opt/minecraft-k8s/monitoring-values.yaml
```

3.3 Apply Alerts & Port-Forward Grafana

Apply the Prometheus rules manifest to register the alerts:

```
sudo k3s kubectl apply -f /opt/minecraft-k8s/monitoring/prometheus-rules.yml
```

To access Grafana without exposing it publicly, establish a secure SSH tunnel from the local WSL machine:

```
ssh -i ~/.ssh/cs312-key.pem -L 3000:localhost:3000 ubuntu@[EC2-IP] \
"sudo KUBECONFIG=/etc/rancher/k3s/k3s.yaml kubectl port-forward -n monitoring
svc/monitoring-grafana 3000:80"
```

Access the dashboard at `http://localhost:3000`.

4. Observability and Dashboards

The custom **Minecraft Server Health** dashboard aggregates critical metrics to answer the operator question: *"Is my service healthy right now?"* at a single glance.

- **Node Health Panels:** Gauges tracking Node Memory Usage %, Node CPU Usage %, and Node Disk Usage % ensure the underlying EC2 instance is not saturated.
- **Pod Health Panels:** Stats tracking Minecraft Pod Restarts and Pod Ready status. To clean up the visualization and hide historical terminated pods, these queries wrap the raw metric in a `sum()` function.
- **Pod Resource Consumption Panels:** Tracks CPU and memory resource consumption specifically for the Minecraft container (using `container_cpu_usage_seconds_total` and `container_memory_working_set_bytes`) to detect memory leaks and CPU throttling before they affect gameplay.
- **Minecraft-Specific Signal:** The dashboard tracks application-level metrics via the `minecraft-exporter` sidecar. Specifically, `minecraft_players_online` visualizes active player counts, and `minecraft_rcon_up` validates active RCON connectivity to the server. Together, these signals verify that the Minecraft application engine is healthy and accepting TCP connections, providing a robust application-layer check compared to basic container running states (`kube_pod_container_status_running`).

5. Alert Runbooks & On-Call Quickstart

On-Call Quickstart: When paged, check the Grafana *Minecraft Server Health* dashboard first. To determine if the outage is user-visible, check the Grafana dashboard's *Pod Ready* and *Service Reachability* panels (or run `nmap -sV -Pn -p T:25565 [IP]`). If the Minecraft port is closed or the pod ready state is 0, players cannot connect. If the pod is not ready, run `kubectl describe pod` to identify scheduling or image pull errors. Check S3 backup logs if disk usage is full.

5.1 Alert: MinecraftPodCrashLooping

Trigger: `increase(kube_pod_container_status_restarts_total...[10m]) >= 5`

Player-Visible Impact: Yes. Players cannot connect or are actively being disconnected repeatedly.

Threshold Justification: 5 restarts within a 10-minute window indicates a genuine crash loop rather than a standard initialization delay.

First-Response & Resolution Steps:

1. Retrieve the logs of the crashed container's previous execution:

```
sudo k3s kubectl logs -l app=minecraft -c minecraft --previous
```

2. Look for JVM initialization errors. If there is a Java version mismatch, update the ConfigMap to pin the correct version:

```
sudo k3s kubectl patch configmap minecraft-config --type=merge -p '{"data":{"VERSION":
"1.20.4"}}'
```

3. Perform a rollout restart to apply the configuration:

```
sudo k3s kubectl rollout restart deployment/minecraft-server
```

5.2 Alert: NodeMemoryPressureHigh

Trigger: Available memory / Total memory * 100 > 85% for 5m.

Player-Visible Impact: Indirectly. May cause server lag or an eventual out-of-memory kill (which occurred to the Grafana pod during setup).

Threshold Justification: 85% provides an early warning buffer, allowing time to scale resources before a hard OOM kill occurs.

First-Response & Resolution Steps:

1. Identify memory-intensive pods on the cluster namespace:

```
sudo k3s kubectl top pods -A
```

2. Identify memory-intensive processes running directly on the host:

```
top -o %MEM -b -n 1 | head -n 20
```

3. If Prometheus or Grafana is consuming excessive memory, update their limits in the monitoring values:

```
sudo KUBECONFIG=/etc/rancher/k3s/k3s.yaml helm upgrade monitoring
prometheus-community/kube-prometheus-stack -n monitoring -f
/opt/minecraft-k8s/monitoring-values.yaml
```

5.3 Alert: NodeDiskUsageHigh

Trigger: Root disk usage is > 80% for 5 minutes.

Player-Visible Impact: Indirectly. World saves and S3 backups will silently fail if disk hits 100%.

Threshold Justification: 80% warns the operator early so they can clear space before world corruption occurs.

First-Response & Resolution Steps:

1. Check host disk usage by directory to locate large files:

```
sudo du -sh /* 2>/dev/null | sort -h
```

2. Prune containerd builder and container image caches:

```
sudo crictl rmi --prune
```

3. Vacuum system journal logs to free space instantly:

```
sudo journalctl --vacuum-time=1d
```

6. Log Investigation & Incident Drill

6.1 Log Investigation

To retrieve raw logs, execute the following command to tail the last 20 lines of the Minecraft server container log:

```
sudo k3s kubectl logs -l app=minecraft -c minecraft --tail=20
```

Look for the line: `[Server thread/INFO]: Done (...s)! For help, type "help"`. This specific event confirms the JVM has finished loading the world and is actively accepting TCP connections.

Log Location & Search Strategy: Logs physically reside on the EC2 host under the containerd runtime directories (specifically at `/var/log/pods/default_minecraft-server-.../*`) and are retrievable via the Kubernetes API. When troubleshooting an incident, the operator should first search for keywords like `WARN`, `ERROR`, `Exception`, or `OutOfMemory`, and verify container initialization by looking for the `Done` startup confirmation.

6.2 Incident Drill: Bad Deployment Rollout

Failure Introduced: An update was deployed utilizing an invalid image tag:

```
sudo k3s kubectl set image deployment/minecraft-server minecraft=...:nonexistent-tag
```

Detection: The `kubectl rollout status` command hung indefinitely. A diagnostic check of the events (`sudo k3s kubectl get events --sort-by=.lastTimestamp`) explicitly highlighted an `ErrImagePull` and subsequent `ImagePullBackOff` error.

Recovery: The deployment was seamlessly reverted to the previous functional ReplicaSet:

```
sudo k3s kubectl rollout undo deployment/minecraft-server
```

Confirmation: Executed `nmap -sV -Pn -p T:25565 [IP]` to confirm the port was open and the MOTD broadcasted correctly.

6.3 Incident Postmortem & Prevention

Root Cause: A bad deploy was executed by setting the Minecraft deployment to a non-existent image tag. This caused Kubernetes to transition the pod to an `ErrImagePull` and `ImagePullBackOff` state, halting the rollout.
Recovery: Executed `kubectl rollout undo` to rollback to the previous stable ReplicaSet.
Prevention: Implement a pre-push image verification process or a validation step in our deployment scripts to verify that image tags exist in the ECR registry before updating the deployment manifest.

7. Cost Controls & Teardown Checklist

- **Data Retention Limitations:** Prometheus metrics retention is explicitly capped to 3 days (`retention: 3d`) in `monitoring-values.yaml` to prevent EBS disk bloat over time.
- **Memory Guardrails:** Strict memory requests (256Mi) and limits (512Mi) are placed on the Prometheus and Grafana pods to prevent them from starving the Minecraft JVM.
- **Automated Stop Schedule:** The EC2 instance is configured with a nightly stop schedule (e.g. stopped daily at 7:00 PM via AWS Instance Scheduler or cron) to prevent idle compute billing when the lab is not in use.
- **Monitoring Stack Teardown:** Run `helm uninstall monitoring -n monitoring` to instantly drop the resource load of the observability stack.
- **Infrastructure Teardown:** Run `terraform destroy` from `ops3-4-5/` on the local machine after recording completes to terminate the EC2 instance and avoid unwanted billing. Note: This teardown is performed outside the video walkthrough itself.